EFFECTIVE CARTIER DIVISORS WITH AMPLE RESTRICTIONS PRODUCE AN AMPLE FAMILY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 23369-23389, immediately following the statement at 23360-23367.

Editorial note. Three cross-chapter pointers (Morphisms, Properties) are recorded in assembly.external_references.

Editorial note. The author's step 'After replacing N by N + 1 we see that s' vanishes at every point of D_i' is retained verbatim; it is the technical point of the proof.

Allowed standard prerequisites (editorial).

- Ampleness of an invertible module via affine non-vanishing loci. Properties, definition properties-definition-ample, cited at proof line 23375. Turns ampleness of L_i restricted to U_i into the existence of n and a section whose non-vanishing locus is affine.
- Extending sections across an effective Cartier divisor. Properties, lemma properties-lemma-invert-s-sections, cited at proof lines 23376-23379. Extends the section from the complement of D_i to all of X after twisting by a high multiple of O_X(D_i).
- Definition of an ample family of invertible modules. Morphisms, definition morphisms-definition-family-ample-invertible-modules, cited at proof lines 23386-23388. Fixes the property that the constructed sections have to exhibit.

Source reference. Original reference 0FXR: focused-seeds/geometry/S9-morph-get-ample-family/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01PS: focused-seeds/geometry/S9-morph-get-ample-family/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01PW: focused-seeds/geometry/S9-morph-get-ample-family/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Suppose given effective Cartier divisors $D_1, \ldots, D_m$ on X and invertible modules $\mathcal{L}_1, \ldots, \mathcal{L}_m$ such that $\bigcap D_i = \emptyset$ and $\mathcal{L}_i|_{X \setminus D_i}$ is ample. Then X has an ample family of invertible modules.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $x \in X$. Choose an index $i \in \{1, \ldots, m\}$ such that $x \notin D_i$. Set $U_i = X \setminus D_i$. Since $\mathcal{L}_i|_{U_i}$ we can find an $n \geq 1$ and a section $s \in \Gamma(U_i, \mathcal{L}_i^{\otimes n})$ such that the locus $(U_i)_s$ where s doesn't vanish is affine (Properties, Definition 01PS). Since U_i is the locus where the canonical section $1 \in \mathcal{O}_X(D_i)$ doesn't vanish, we see from Properties, Lemma 01PW there exists an $N \geq 0$ such that s extends to a section

$$s' \in \Gamma(X, \mathcal{L}_i^{\otimes n} \otimes_{\mathcal{O}_X} \mathcal{O}_X(ND_i))$$

After replacing N by $N + 1$ we see that s' vanishes at every point of D_i and hence that $X_{s'} = (U_i)_s$ is affine. This proves that X has an ample family of invertible modules, see Morphisms, Definition 0FXR. $\square$